Microcontrollers, flowcharts & programming

ENGINEERING WITH MR KELLY

National 5 · Electronics & control · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ draw flowcharts with start/stop, input, output, decision and loop symbols
- ☐ use fixed loops, continuous loops and conditional loops
- ☐ write programs using high, low, pause, if…then, for…next

I DO — Worked example

Read how a full-mark answer is built

A flowchart switches on an LED on output 2 when a switch on input 0 is pressed, waits 3 seconds, then switches it off, and repeats.

1 Write a program for the flowchart. (4 marks)

Points that earn marks from the marking instructions: one ✓ per point

- ✓ main: label and if pin0 = 1 then (or input check loop)
- ✓ high 2
- ✓ pause 3000
- ✓ low 2 and goto main

WE DO — Guided practice

Use the starters and word bank to help you

A traffic light sequence must repeat 20 times then stop.

2 Describe how a fixed loop is used in the program. (2 marks)

Sentence starters

First, …

Next, … using a …

Word bank

sequence · program · surrounds · repeats

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) In PBASIC, pause 2000 waits for...
☐ 2 ms ☐ 0.2 s ☐ 20 s ☐ 2 s

b) Which symbol tests whether a switch is pressed?
☐ Parallelogram ☐ Diamond ☐ Rectangle ☐ Oval

c) An advantage of a microcontroller over a hard-wired circuit is...
☐ It can be reprogrammed without changing the circuit ☐ It never needs a power supply to run ☐ It works without any input devices ☐ It is always faster than a hard-wired circuit

d) Which is an advantage of a microcontroller over a hard-wired circuit?
☐ It needs no software to be written ☐ Fewer components, so more reliable ☐ It is always cheaper than any chip ☐ It never needs a driver for outputs

4 Key terms: write the letter of the matching description next to each term.

1. **Decision symbol**	____	**A** Diamond with Y/N exits, used to test an input or a count.
2. **Output symbol**	____	**B** Repeats a set number of times.
3. **Delay symbol**	____	**C** Rectangle with a time and unit, e.g. "wait 2 s".
4. **Fixed loop**	____	**D** Repeats forever.
5. **Continuous loop**	____	**E** Parallelogram, e.g. "pin 7 on".

A flowchart uses different symbols.

5 Draw the flowchart symbol for a decision and for an output. (2 marks)

6 Explain what is meant by ‘pause 500 (PBASIC)’. (2 marks)

7 Describe the advantage of a microcontroller. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?